

Process Characterization Report

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCR-005

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCR-005
Document class	Process Characterization Report
Title	Process Characterization Report: Protein A Chromatography (Step 5)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

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1 Results

1.1 Response-surface models

The response-surface stage produced one model that is adequate and predictive, one that is adequate, and one that is neither. The fit of all three is given in Table 1.1.

Table 1.1: Response-surface model fit for the three responses.

Response	N	R ²	Adj. R ²	Pred. R ²	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.989	0.976	0.932	80.9	2.71e-10	895
Step yield	28	0.913	0.818	0.586	9.69	0.000105	0.016
Leached Protein A (ppm)	28	0.567	0.101	-0.714	1.22	0.365	0.637

Pool host cell protein is modelled well. The full quadratic model reaches $R^2 = 0.989$, adjusted $R^2 = 0.976$ and predicted $R^2 = 0.932$, with a model F of 80.9 ($p = 2.7e-10$). A predicted coefficient close to the adjusted coefficient means the model predicts a held-out run about as well as it fits the data used to build the model, which is the property the design space in §6 depends on. Step yield is adequate but less predictive, at $R^2 = 0.913$ and predicted $R^2 = 0.586$, which is enough to describe the yield surface and to support the classification of the two key process parameters in §9.

Leached Protein A yielded no usable model. The quadratic fit reaches $R^2 = 0.567$ (Table 1.1), but the adjusted coefficient falls to 0.101 and the predicted coefficient is -0.714. A negative predicted coefficient means the model predicts a held-out run worse than the mean of the data predicts. The model F ratio is 1.22 at $p = 0.36$, which is to say that the model as a whole explains no more than chance would. This report therefore makes no predictive use of it, with the treatment of the attribute set out below.

The coefficient table for pool host cell protein is given in Table 1.2, in which 7 of the 14 terms reach significance at $\alpha = 0.05$.

Table 1.2: Full quadratic coefficients for pool host cell protein, on coded factors.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	7,215.5	309	23.3	5.44e-12	***
A	3,871.8	211	18.4	1.12e-10	***
B	-5,244.6	211	-24.9	2.4e-12	***

Term	Coef.	Std. err.	t	p-value	Sig.
C	-435.4	211	-2.06	0.0596	
D	-1,055.5	211	-5	0.000242	***
AB	-1,899.8	224	-8.49	1.16e-06	***
AC	-429.1	224	-1.92	0.0774	
AD	-518.1	224	-2.32	0.0376	*
BC	348.2	224	1.56	0.144	
BD	727.4	224	3.25	0.00632	**
CD	-128.7	224	-0.575	0.575	
A ²	-344.5	557	-0.618	0.547	
B ²	1,943.5	557	3.49	0.00401	**
C ²	851.3	557	1.53	0.151	
D ²	330.5	557	0.593	0.563	

Elution buffer pH carries the largest coefficient at -5,245 ng/mg per coded unit, with protein load next at 3,872. The interaction of the two is significant at -1,900, reproducing the sign and roughly the magnitude estimated in the screening stage. The quadratic term in elution pH is significant at 1,944 ($p = 0.0040$), which is the curvature a two-level design cannot see. End of pool collect reaches significance in this model at -1,055 ng/mg per coded unit, whereas the screening estimate for the same factor was not significant and carried the opposite sign (Table 1.4). The two estimates do not conflict, because a non-significant screening estimate makes no claim about the sign. Therefore the response-surface estimate governs, in line with the framing set out in §5.2.

The residual for pool host cell protein is partitioned into lack of fit and pure error in Table 1.3.

Table 1.3: Analysis of variance for pool host cell protein, with the residual partitioned against the centre-point pure error.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	9.08e+08	14	6.48e+07	80.9	2.71e-10
Residual	1.04e+07	13	8.01e+05	nan	nan
Lack of fit	1e+07	10	1e+06	7.88	0.058
Pure error	3.82e+05	3	1.27e+05	nan	nan

Lack of fit does not reach significance at $\alpha = 0.05$ ($F = 7.88$, $p = 0.058$), which supports the quadratic form for this response. However, that p-value is the closest to α of the three responses, on a test that rests on 4 centre-point replicates. For this reason the result bounds the evidence for the model form without establishing it. The design space in §6 is bounded accordingly. The corresponding tests for step yield and leached Protein A are further from significance, at $p = 0.73$ and $p = 0.95$ (Appendix C).

The diagnostics for pool host cell protein are given in Figure 1.1, where the residuals show no structure against the predicted value, the normal plot is close to a straight line and the actual values fall on the identity line across the whole range of the response.

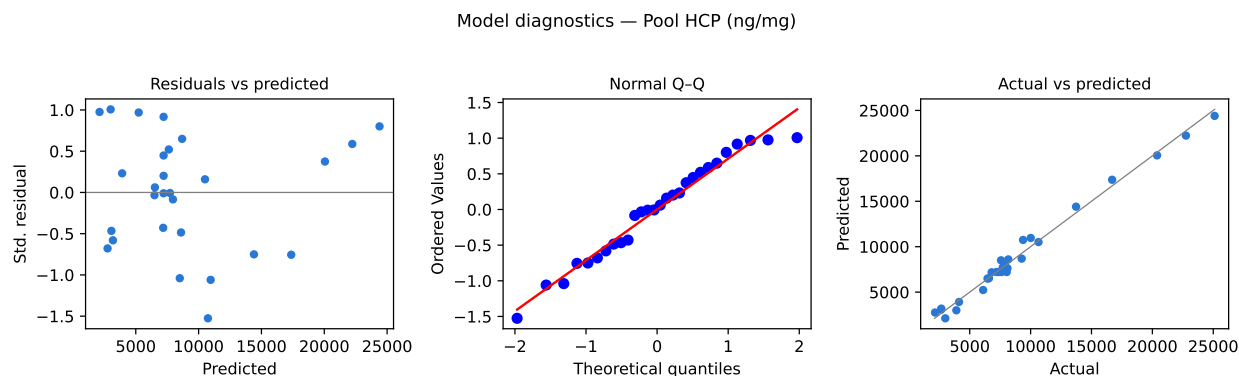


Figure 1.1: Model diagnostics for pool host cell protein: standardized residuals against predicted value, normal quantile plot, and actual against predicted.

The fitted surfaces are shown in Figure 1.2 over protein load and elution buffer pH, with the other two factors held at the design centre. Pool host cell protein rises steeply towards the corner at high load and low pH. The contours are curved instead of parallel, which is the interaction and the pH curvature already seen in Table 1.2. Step yield varies over a narrow band across the same plane, and leached Protein A shows almost no gradient, consistent with a model that explains nothing.

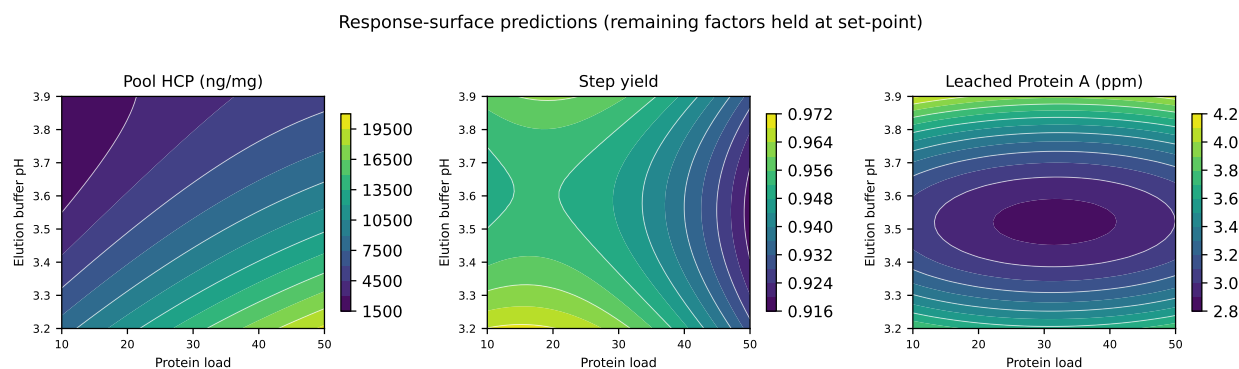


Figure 1.2: Response-surface predictions for the three responses over protein load and elution buffer pH, with load flow rate and end of pool collect held at the design centre.

The leached Protein A model is retained as knowledge-space evidence and is put to no other use in this report. The diagnostics for that model are given in Figure 1.3. Exactly one of the 14 terms reaches significance, the quadratic in elution pH at $p = 0.040$, which is about what testing that many terms at $\alpha = 0.05$ produces by chance. Lack of fit is far from significant ($p = 0.95$, Appendix C). That test compares the residual with the pure error. The pure error for this response is large (33.8 % at the centre point). A non-significant result here therefore says only that the quadratic form is not contradicted by a noisy replicate set, which is not evidence that the model predicts.

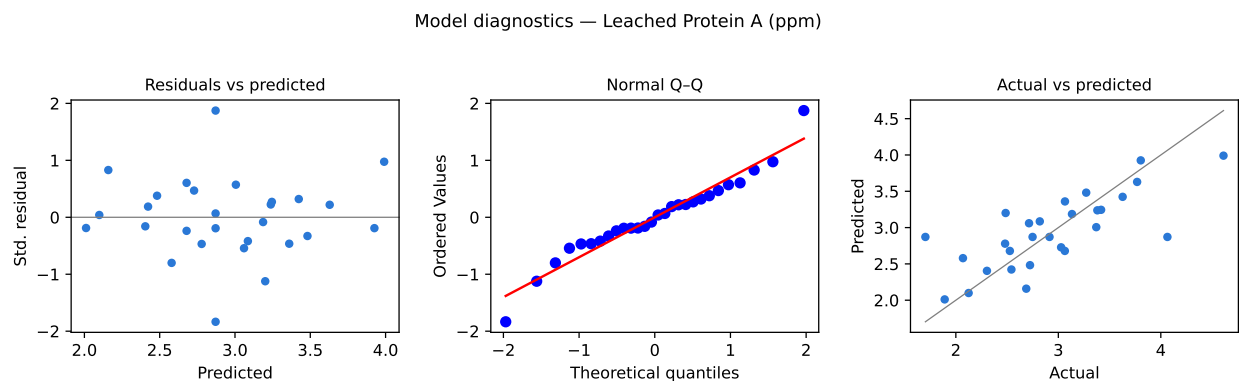


Figure 1.3: Model diagnostics for leached Protein A. The actual against predicted panel shows the scatter that a predicted R-squared below zero describes.

The operative claim for leached Protein A is therefore made from the assayed pools directly. Across the 47 runs of the two designs, the pools ranged from 1.70 to 4.61 ppm, with a mean of 2.95 ppm and a standard deviation of 0.58 ppm, against an acceptance criterion of 5 ppm, which 47 of them met. The highest single pool left 0.39 ppm of margin, which is narrow on one run and is the reason the attribute is monitored on every batch (\$10). However, the criterion applied here is the drug-substance specification, which is applied to the capture pool conservatively. Leached Protein A is cleared downstream of this step by cation exchange and anion exchange (PCR-007, PCR-008). In addition, only 0.003 ppm of leached Protein A is carried by the simulated drug substance at commercial scale (\$8).

1.2 Mechanistic interpretation

The surfaces in Figure 1.2 follow from the physical chemistry of affinity capture and confirm the expectations recorded in §2.1. Elution buffer pH governs pool host cell protein because it sets the aggressiveness of desorption: a lower pH releases the antibody faster and also releases the more tightly associated host cell protein, which then co-elutes into the collected pool. The convex quadratic term means the effect is not uniform across the range: host cell protein falls steeply as pH rises from the low edge and then flattens above the set-point, which leaves the pool most sensitive to pH exactly where the pH is lowest.

Protein load acts through the capacity of the bed: at a light load the antibody occupies a small fraction of the available ligand and the impurity that binds non-specifically is washed away efficiently, whereas a load approaching dynamic binding capacity extends the mass-transfer zone and retains a greater share of the loaded impurity into the elution. The negative interaction between load and pH follows directly: at a high elution pH part of the impurity recruited by a heavy load is left behind on the resin in any case, which makes the two effects less than additive over the characterized ranges.

Step yield is governed by where the pool is stopped. Collecting further down the descending edge recovers antibody that would otherwise go to strip, which is why end of pool collect carries the dominant positive term. By contrast, load and flow act in the opposite direction through breakthrough during the load phase. The interaction between them is negative (Table 1.5) because a short residence time matters most when the bed is already close to saturation.

Leaching of the affinity ligand behaves as a resin property over these ranges, since ligand release depends on the chemistry of the immobilization, on the cumulative cycle count and on the sanitization history, none of which is a parameter of a single run. Nothing in the data contradicts that account. The four operating parameters together explain no more of the pool variation than chance would, while the centre-point replicates scatter more widely than any factor moved the response.

These results also explain why a multivariate region is required in place of four independent ranges. The two parameters that bound the operating region interact, with the surface curved in one of them. A univariate range for protein load would have to be set at the load that is acceptable at the worst elution pH. A univariate range for elution pH would have to be set at the pH that is acceptable at the heaviest load. Consequently, both would be narrower than the process needs, whereas the multivariate region in §6 states the constraint the process actually faces.

Appendix — screening effect tables referenced in the text

Table 1.4: Screening effect estimates for pool host cell protein, the six largest by magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	-8,840.3	-4,420.2	350	-12.6	1.45e-06	***
A	7,269.4	3,634.7	350	10.4	6.4e-06	***
AB	-2,586.7	-1,293.4	350	-3.7	0.00608	**
AD	1,132.9	566.4	350	1.62	0.144	
AC	830.1	415.1	350	1.19	0.27	
D	707.9	353.9	350	1.01	0.341	

Table 1.5: Screening effect estimates for step yield, the six largest by magnitude.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
D	0.0604	0.0302	0.00208	14.5	5.01e-07	***
A	-0.0255	-0.0128	0.00208	-6.13	0.00028	***
C	-0.0237	-0.0118	0.00208	-5.68	0.000463	***
AC	-0.0209	-0.0104	0.00208	-5.01	0.00104	**
CD	-0.0107	-0.00537	0.00208	-2.58	0.0326	*
BD	-0.00981	-0.0049	0.00208	-2.36	0.0463	*